

2.5 Statistical Hypothesis Testing

Mathematics B (MEI) — OCR B — A-Level (H640) — Statistics — spec refs H1–H11

What this topic covers. The language and logic of hypothesis testing, tests on a binomial probability, the distribution of the sample mean and tests on a population mean using the Normal distribution, and using a given correlation coefficient to make an inference.

1. The language

Term	Meaning
Null hypothesis H_0	The default assumption being tested — always an equality.
Alternative hypothesis H_1	What is concluded if H_0 is rejected.
Test statistic	The quantity calculated from the sample and compared against the critical value.
Significance level	The probability of incorrectly rejecting H_0 .
Critical region	The values of the test statistic that lead to rejection. Also called the rejection region.
Acceptance region	The values for which H_0 is not rejected.
Critical value	The boundary of the critical region.
p-value	The probability of a result at least as extreme as the one observed, assuming H_0 is true.

Hypotheses are always statements about a **population parameter** — p or μ — never about the sample. Writing $H_0 : \bar{x} = 50$ is wrong; it must be $H_0 : \mu = 50$.

2. One-tailed or two-tailed?

Wording	Test	H_1
"biased towards heads", "has increased"	One-tailed	$p > 0.5$
"has decreased", "is less effective"	One-tailed	$p < 0.5$
"is biased", "has changed"	Two-tailed	$p \neq 0.5$

For a two-tailed test at the α level, the probability is split between the tails — so a 5% two-tailed test uses 2.5% in each tail.

The word that decides this is whether a **direction** is specified. "Biased" alone gives no direction and demands a two-tailed test; "biased towards heads" gives one.

3. The logic of the test

1. State H_0 and H_1 in terms of a population parameter.
2. Assume H_0 is true.
3. Calculate how likely the observed result — or something more extreme — would be under that assumption.
4. If that probability is below the significance level, the result is unlikely under H_0 , so **reject** H_0 .
5. If not, there is **insufficient evidence** to reject H_0 .
6. State the conclusion in the **context** of the original question.

Never write "accept H_0 ". Failing to reject is not proof of truth — it means the evidence was not strong enough. Similarly, rejecting H_0 gives **evidence**, never proof. Both phrasings are penalised.

The significance level is the probability of **incorrectly rejecting** a true null hypothesis (spec ref H3). A 5% level means that, if H_0 is true, there is at most a 5% chance of wrongly rejecting it.

4. Binomial hypothesis tests

Here H_0 takes the form $p =$ a particular value, and the test statistic is the observed number of successes (H4–H6).

Worked example. A coin is suspected of being biased towards heads. Set up the test.

$$H_0 : p = 0.5, \quad H_1 : p > 0.5$$

One-tailed, since a direction is specified. Reject H_0 if the probability of getting at least the observed number of heads is below the significance level.

Worked example. A test at the 5% level gives a p-value of 0.032. State the conclusion.

Since $0.032 < 0.05$, reject H_0 . There is evidence at the 5% level to support the alternative hypothesis, stated in context.

Worked example. The same test gives a p-value of 0.08.

Since $0.08 > 0.05$, do not reject H_0 . There is insufficient evidence at the 5% level.

Because the binomial distribution is **discrete**, the probability of landing in the rejection region is usually *less* than the intended significance level, and only rarely equal to it. MEI notes this explicitly and states that learners are not tested on the distinction — but it explains why critical regions look slightly "off".

5. The distribution of the sample mean

For random samples of size n drawn from $N(\mu, \sigma^2)$, the sample mean is itself Normally distributed (H7):

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

so its standard deviation is $\frac{\sigma}{\sqrt{n}}$.

For $X \sim N(100, 36)$ and samples of size 9, the sample mean has variance $\frac{36}{9} = 4$ and standard deviation 2.

The **Central Limit Theorem** extends this: for a large sample, the sample mean is approximately Normal *even when the population is not*. It is the reason a Normal test on a mean is so widely used in practice.

Not examined on H640. The Central Limit Theorem is in the exclusions column for H7, so never quote it as your justification. What H640 asks you to know is the *stated* result above — that a random sample drawn from $N(\mu, \sigma^2)$ has $\bar{X} \sim N(\mu, \sigma^2/n)$ — together with the H8 conditions (population variance known, or unknown with a large sample). The CLT is kept here as background for why those conditions are reasonable, not as something to revise.

Notice that increasing n shrinks $\frac{\sigma}{\sqrt{n}}$. A larger sample makes the sample mean cluster more tightly around μ , so genuine differences become easier to detect.

6. Testing a population mean

Valid when the population variance is known, or is unknown but the sample is large (H8). The significance level will be given.

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Worked example. A sample of 25 from $N(\mu, 100)$ has mean 44. Test $H_0 : \mu = 40$ against $H_1 : \mu > 40$ at the 5% level.

Under H_0 , $\bar{X} \sim N(40, \frac{100}{25}) = N(40, 4)$, so the standard deviation is 2.

$$z = \frac{44 - 40}{2} = 2$$

The one-tailed 5% critical value is 1.645. Since $2 > 1.645$, the result lies in the critical region: reject H_0 . There is evidence at the 5% level that the population mean exceeds 40.

A Normal test on a mean is **not** valid with a small sample and an unknown population variance — with $n = 8$ and σ unknown, neither condition is met.

7. Correlation

Correlation measures how close data points lie to a straight line; a **rank** correlation coefficient measures it between the ranks rather than the raw values (H10).

For a correlation test, the null hypothesis is that there is **no correlation** (or no association) in the population. You compare a *given* coefficient against a given critical value or p-value and draw the inference (H11).

What is excluded. Calculating a correlation coefficient is not required, and neither is knowing the names of particular coefficients. Questions supply the coefficient and indicate whether correlation or association is being tested. "Association" refers to a more general relationship than a straight-line one.

A significant correlation still does not establish **causation**. A third variable may drive both — the standard example being hot weather driving both ice cream sales and drownings.

8. Interpreting results honestly

- The significance level must be chosen **before** the data are examined.
- Lowering the level from 5% to 1% demands stronger evidence and makes rejection harder.
- A result significant at 5% but not at 1% has a p-value between 0.01 and 0.05.

- Running many tests at the 5% level on data with no real effect will still produce false positives — about 1 in every 20.

9. Common pitfalls

- Writing hypotheses about the sample statistic rather than the population parameter.
- Using a one-tailed test when no direction is specified.
- Forgetting to halve the significance level for each tail in a two-tailed test.
- Writing "accept H_0 " or claiming something is "proved".
- Dividing by σ instead of $\frac{\sigma}{\sqrt{n}}$ when testing a mean.
- Omitting the conclusion in context.
- Using a Normal test with a small sample and unknown variance.
- Choosing the significance level after seeing the result.